

Exercise 2: E-commerce Platform Search Function

Objective

Implement a simple E-commerce Product Search using Linear Search in C#.

Concepts Learned

- Classes and Objects
- List Collection
- Linear Search Algorithm
- foreach Loop
- if Condition
- User Input using Console.ReadLine()
- Object-Oriented Programming (OOP)

Algorithm

1. Create a Product class.
2. Store products in a List.
3. Take Product ID as input from the user.
4. Traverse the list using Linear Search.
5. If Product ID matches, display product details.
6. Otherwise display "Product Not Found".

Output

- Displays product details when Product ID exists.
- Displays "Product Not Found" for an invalid Product ID.

Time Complexity

- Best Case: $O(1)$
- Worst Case: $O(n)$